# 21ECC322L- Communication Laboratory

*Academic Year 2025-26*

**Mini Project Proposal Form**

*(Department of Electronics and Communication Engineering)*

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**1.Project Title:** Design and Implementation of a WI-FI JAMMER using ESP32

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## 2. Team Members (with Reg.Nos):


Amit Chauhan - RA2311004010332

Akrity Madhukar- RA2311004010341

Swayam Mishra - RA2311004010349

Lakshita - RA2311004010354


## 3. Project Category:

- Digital Communication
- Microwave Communication

## 4. Project Objective:

The objective of this project is to design and implement a Wi-Fi jammer using the ESP32 microcontroller that can disrupt 2.4 GHz Wi-Fi networks and also analyze nearby Wi-Fi networks by monitoring packet transmission. The system will include an OLED display for network information and push buttons for user control of different modes such as scanning, jamming, and packet monitoring.

## 5. Problem Statement:

Wi-Fi networks operating in the 2.4 GHz band are widely used for wireless communication. These networks are susceptible to interference and packet flooding attacks which can disrupt communication. Network administrators and researchers often need tools to analyze network traffic, packet transmission rates, and network vulnerabilities.

The problem addressed in this project is to develop a portable device using ESP32 that can:

- Scan nearby Wi-Fi networks
- Monitor packet traffic
- Perform packet flooding/jamming in the 2.4 GHz band

- Display network information on an OLED screen

This project demonstrates wireless network interference, packet analysis, and IoT-based network monitoring.

## 6. Literature Review (Brief Summary):

ESP32 is a low-cost microcontroller with built-in Wi-Fi and Bluetooth capabilities operating in the 2.4 GHz band. It supports different Wi-Fi modes such as Station mode, Access Point mode, and Promiscuous mode. Promiscuous mode allows the ESP32 to capture Wi-Fi packets without connecting to a network, making it useful for network analysis and packet monitoring.

Wi-Fi jamming using ESP32 is commonly implemented using deauthentication packet attacks, where fake deauthentication packets are sent to devices connected to a Wi-Fi network, forcing them to disconnect from the network.

Previous studies and projects show that ESP32 can be used for:

- Wi-Fi packet sniffing
- Network scanning
- Deauthentication attacks
- Signal strength monitoring
- Packet traffic analysis

This project combines packet analysis and Wi-Fi jamming into a single embedded system device with OLED display and push button interface.

## Step 1 – Hardware Setup

Components used:

- ESP32 microcontroller
- 0.96-inch OLED display (I2C)
- Breadboard
- Jumper wires
- Push buttons (4)
- USB power supply

## Step 2 – Wi-Fi Network Scanning

ESP32 scans nearby Wi-Fi networks and displays:

- Network name (SSID)
- Signal strength (RSSI)
- Channel number
  on the OLED display.

## Step 3 – Packet Monitoring

ESP32 operates in **promiscuous mode** to capture Wi-Fi packets and count how many packets are being transmitted in the network. The packet count and traffic rate will be displayed on the OLED screen.

## Step 4 – Wi-Fi Jamming

ESP32 sends **deauthentication packets** to devices connected to a selected Wi-Fi network, causing them to disconnect from the network. This effectively jams the Wi-Fi communication in the 2.4 GHz band.

## Step 5 – Push Button Control

Push buttons are used for:

- Button 1 → Scan Networks
- Button 2 → Select Network
- Button 3 → Start Packet Analysis
- Button 4 → Start/Stop Jamming

## Step 6 – Display Output

OLED display will show:

- Available networks
- Selected network
- Packet count
- Jamming status
- Channel information

## 7. Fabrication & Prototyping

**PCB/Waveguide Design:** No
**Fabrication Method:** Breadboard Prototyping

## 7.3 Testing & Validation

Testing will be done by:

- Scanning available Wi-Fi networks
- Monitoring packet transmission count
- Checking device disconnection during jamming
- Measuring signal strength before and after jamming

**Equipment Used:**

- ESP32
- OLED Display
- Wi-Fi Router
- Laptop / Mobile Wi-Fi Analyzer
- Serial Monitor

## 8. Expected Outcome & Deliverables

Expected outcomes of the project:

- Portable Wi-Fi jammer device using ESP32
- Wi-Fi network scanner
- Packet traffic analyzer
- OLED display interface
- Push button control system
- Demonstration of Wi-Fi deauthentication attack
- Mini project report and documentation

The system will be able to:

- Detect nearby Wi-Fi networks
- Analyze packet traffic
- Jam selected 2.4 GHz Wi-Fi network
- Display all information on OLED screen

## 9. Resources Required

### Hardware

- ESP32
- 0.96 OLED Display (I2C)
- Breadboard
- Jumper Wires
- Push Buttons (4)
- Resistors
- USB Cable / Power Supply

### Software

- Arduino IDE
- ESP32 Board Package
- Wi-Fi Libraries
- OLED Display Library (SSD1306)
- Serial Monitor

## 12. References


1. ESP32 Technical Reference Manual
2. IEEE 802.11 Wi-Fi Standards
3. ESP32 Wi-Fi Promiscuous Mode Documentation
4. Embedded Systems with ESP32 – IoT Applications
5. Wireless Communication and Networking – William Stallings


## Approval Section

*(For Faculty Use Only)*

| Faculty Name | Approval (✓/✗) | Comments | Date |
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